

Public datasets	PeeringDB, EPA/USGS Water Quality Portal, USGS PAD-US, EIA-861, FEMA National Risk Index. 6 concepts served, 1 proxy retired.
Provenance	2,098 of 2,182 evidence records 96% REAL . The remaining 84 are labelled synthetic documents.
Ingestion	PDF text plus OCR for scans. A transcribed value is carried at half confidence and flagged for a human read: a misread digit is a fabricated number with a citation on it.
Verification	273 tests, lint and typecheck clean.

05
FINDINGS

Three things the system told us

- HARBOUR POINT, VA
Groundwater at **6,610 mg/L** total dissolved solids. Brackish. That changes the cooling design, and it was invisible while the number was synthetic.
- RIO VERDE MESA, AZ
Sits **inside Tonto National Forest** and is rated **Very High** for wildfire. Two independent public datasets agree. An earlier version of our own code reported it as 4 km from a shooting range; we found and fixed the query bug that hid it.
- A SCALE TRAP
FEMA's wildfire index runs 0-100 and looks comparable to any other 0-100 index. It is not: the median tract FEMA calls *Very Low* scores **43.6**. Evenly-spread thresholds would have penalised every safe site in the country by half its wildfire points. Thresholds are now anchored to FEMA's own published class boundaries, with a test that fails if a future release moves them.

06
IMPACT

Why it matters

A single hyperscale campus is a **\$1-3 bn** commitment, decided in weeks on the thinnest evidence in the project's life.

Before construction: the cost of choosing wrong is not the land. It is discovering brackish water, an unbuildable interconnection queue or a wildfire exposure after the option is signed. This surfaces them as tracked gaps with named owners, in hours rather than a six-week study.

During construction: substitutions are constant and each is a small coordination risk. The system flags where structural, electrical or mechanical review is *required*, and never claims adequacy. That distinction is what makes it usable by the engineer who carries the liability.

For Mireye: a demonstration of what the API is worth when something reasons over it rather than displaying it. Every call is credit-metered, cached against the catalog's own TTLs and budget-capped.

07
NEXT

What we do from here

- 1 Replace the last synthetic data.** 84 document records remain; the path is written up. AHRI certified ratings and manufacturer submittals, through the upload flow that already works.
- 2 Four more public datasets.** WRI Aqueduct, USGS 3DEP, USDA CropScape and NOAA ISD, each with its blocker and its procedure documented.
- 3 Production architecture.** Kafka, Redis, Postgres and a projection-based impact graph, with a migration order whose first four steps carry a pilot on their own.

COLOPHON

React 18 · TypeScript · Vite · Tailwind · FastAPI · Pydantic v2 · LangGraph · Pint · Neo4j
AuraDB · SQLite / pgvector · Tesseract · OpenAI · Render (Docker) + Vercel

Repository github.com/RishiiGamer2201/the-wet-stack-mireye, 42 commits, ~21,000 lines, 15 documents including a verified Mireye contract reference, a public-dataset guide and a production low-level design.